

Periodic Table of the Elements

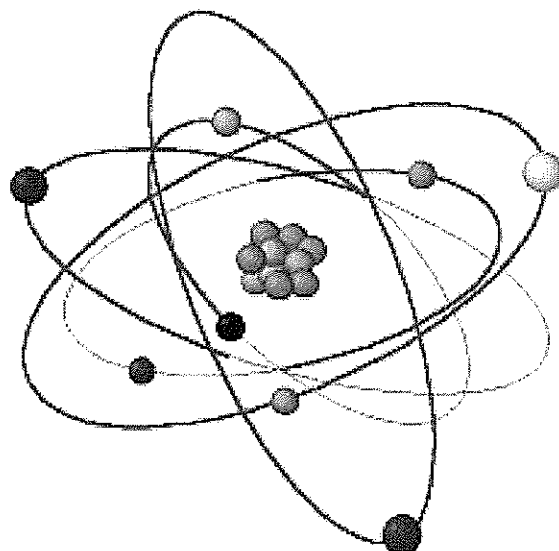
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| 1   | 2   |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     | 18 |
| 3   | 4   |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     | 19 |
| 5   | 6   | 7   | 8   | 9   | 10  | 11  | 12  | 13  | 14  | 15  | 16  | 17  | 18  |     |     |     |     |    |
| 19  | 20  | 21  | 22  | 23  | 24  | 25  | 26  | 27  | 28  | 29  | 30  | 31  | 32  | 33  | 34  | 35  | 36  |    |
| 37  | 38  | 39  | 40  | 41  | 42  | 43  | 44  | 45  | 46  | 47  | 48  | 49  | 50  | 51  | 52  | 53  | 54  |    |
| 55  | 56  | 57  | 58  | 59  | 60  | 61  | 62  | 63  | 64  | 65  | 66  | 67  | 68  | 69  | 70  | 71  | 72  |    |
| 73  | 74  | 75  | 76  | 77  | 78  | 79  | 80  | 81  | 82  | 83  | 84  | 85  | 86  | 87  | 88  | 89  | 90  |    |
| 91  | 92  | 93  | 94  | 95  | 96  | 97  | 98  | 99  | 100 | 101 | 102 | 103 | 104 | 105 | 106 | 107 | 108 |    |
| 109 | 110 | 111 | 112 | 113 | 114 | 115 | 116 | 117 | 118 | 119 | 120 | 121 | 122 | 123 | 124 | 125 | 126 |    |

Lanthanide Series

|    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Ce | Pr | Nd | Pm | Sm | Eu | Gd | Tb | Dy | Ho | Er | Tm | Yb | Lu |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|

Actinide Series

|    |    |   |    |    |    |    |    |    |    |    |    |    |    |
|----|----|---|----|----|----|----|----|----|----|----|----|----|----|
| Th | Pa | U | Np | Pu | Am | Cm | Bk | Cf | Es | Fm | Md | No | Lr |
|----|----|---|----|----|----|----|----|----|----|----|----|----|----|

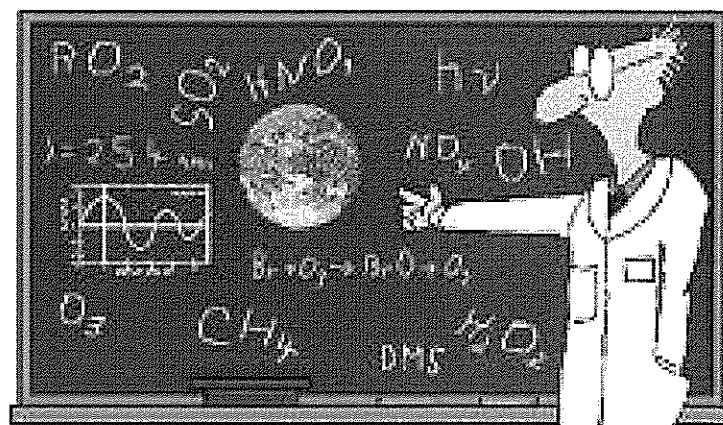
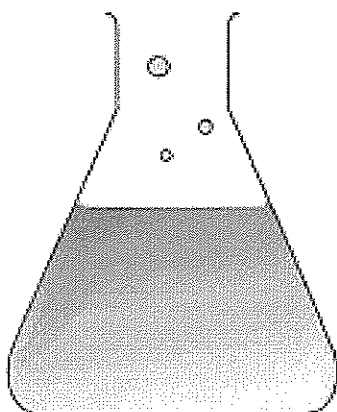


## YEAR 10 SCIENCE

### NATURAL & PROCESSED MATERIALS

### CHEMISTRY

Name: \_\_\_\_\_



# FORMATION OF IONS

## Definitions

***Ion***: an atom that has a charge due to the gain or loss of electrons.

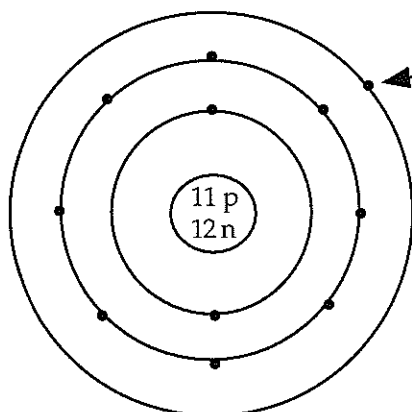
***Valency***: the charge on an ion.

## Positive Ions

- Formed when atoms ***lose electrons***.
- Metal atoms tend to lose electrons from their outer shell.
- Gain a positive charge since there are more protons than electrons.

e.g.

$^{23}_{11}\text{Na}$



Sodium has one electron in its outer shell. If this electron is "lost", the resulting sodium ion will have a +1 charge.

(2, 8, 1)

When the sodium ion forms, it has a full outer shell of eight electrons like the nearest inert gas - neon - which is very stable.

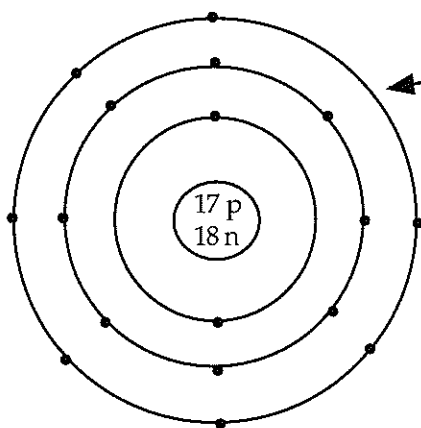
The symbol is written as  **$\text{Na}^{1+}$** .

## Negative Ions

- Formed when atoms *gain electrons*.
- Non-metal atoms tend to gain electrons.
- Gain a negative charge since there are more electrons than protons.

e.g.

$^{35}_{17}\text{Cl}$



Chlorine has a space for one electron in its outer shell. If an electron is "gained", the resulting chlorine ion will have a -1 charge.

(2, 8, 7)

By gaining one electron, the chlorine ion has the same completely full outer shell as the nearest inert gas - argon.

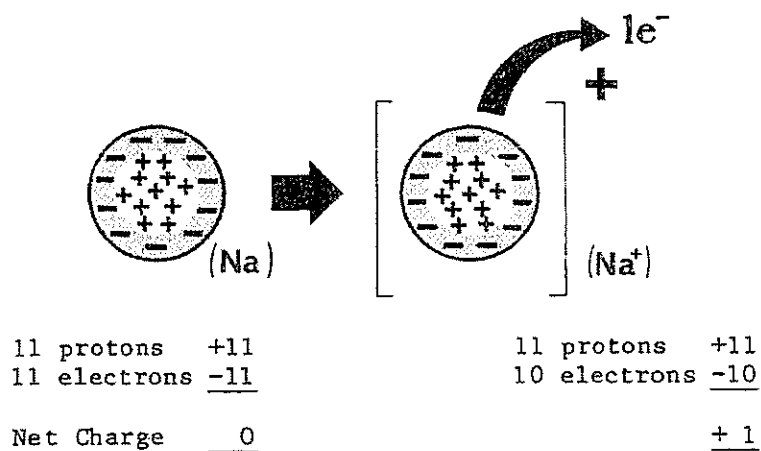
The symbol is written as  $\text{Cl}^{1-}$ .

**You will be given a list of common ions that you can use to name and write the formula of various chemicals.**

# FORMATION OF IONS

## FORMATION OF POSITIVE IONS

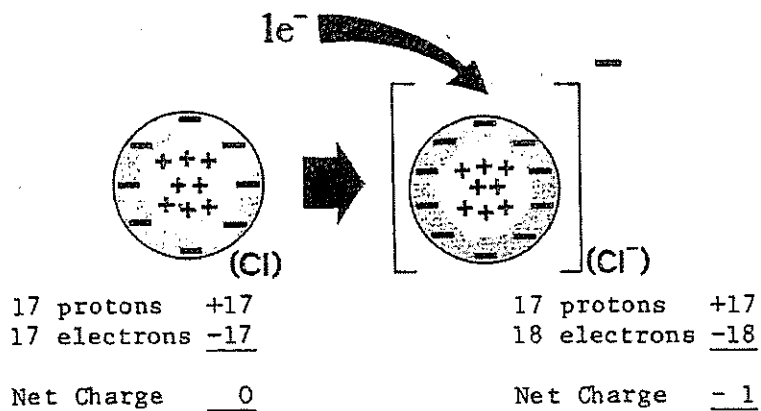
Positive ions are formed when a neutral atom loses one, two or three electrons leaving the ion with more protons than electrons.



CHEMISTS HAVE FOUND THAT METAL ATOMS FORM POSITIVE IONS

## FORMATION OF NEGATIVE IONS

Negative ions are formed when a neutral atom gains one, two or three electrons leaving the ion with less protons than electrons.



CHEMISTS HAVE FOUND THAT NON-METAL ATOMS FORM NEGATIVE IONS

## TYPES OF BONDING

All chemical reactions involve electrons from the reactants interacting with each other to produce more stable electron configurations. In most instances, after elements have reacted, each of the atoms within the molecule has achieved the electron configurations of the noble or inert gases.

The inert gases of Group VIII are very stable due to their filled valence shells. All elements would like to obtain the same electron configuration when bonding - such an arrangement is called a **Stable Octet**.

**Octet Rule:** most elements attain an octet configuration when they bond.

In most cases this means that after reacting, the atoms have eight electrons in their outer shell.

Hydrogen is an important exception to the octet rule, as it achieves only a maximum of two electrons, which is the noble gas configuration of helium.

The two ways in which atoms can achieve noble gas configurations are by:

- *sharing electrons between atoms (covalent bonds).*
- *transferring electrons between atoms (ionic bonds).*

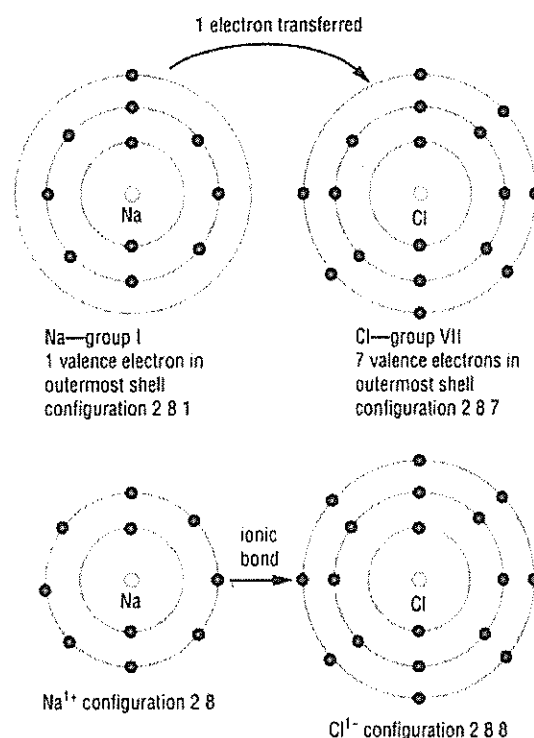
### Ionic Bond

**Ionic Bond:** the electrostatic attraction between oppositely charged ions.

Generally, *all metals plus the ammonium ion  $\text{NH}_4^{1+}$*  form ionic bonds with non-metals from the right hand side of the periodic table.

e.g. Consider the formation of NaCl.

- Na is in Group I and will donate 1 electron. It will now have the same electron configuration as the nearest inert gas Ne. It will form  $\text{Na}^{1+}$  ion.
- Cl is in Group VII and has room for one electron to fill its valence shell like the nearest inert gas Ar. It will form  $\text{Cl}^{1-}$  ion.



This process is described by *electron dot diagrams*, which are used to represent the electrons in the *valence shell*.

e.g.  $\text{Na} \cdot$  represents a sodium atom with one electron in its valence shell.

$[\text{Na}]^{1+}$  represents a sodium ion formed when the one electron is donated from the valence shell.



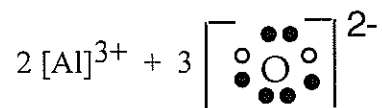
represents a chlorine ion with a donated electron to fill the valence shell.

1 donated electron  
gives a -1 charge on  
the ion.

**Note** *Square brackets [ ] are used for ionic substances.*

### Other Examples

- $\text{Al}_2\text{O}_3$  - show the number of ions involved.



- Draw the electron dot diagrams for the following ions.



## Covalent Bond

**Covalent Bond:** mutual electrostatic attraction for a shared pair of electrons by two adjacent nuclei.

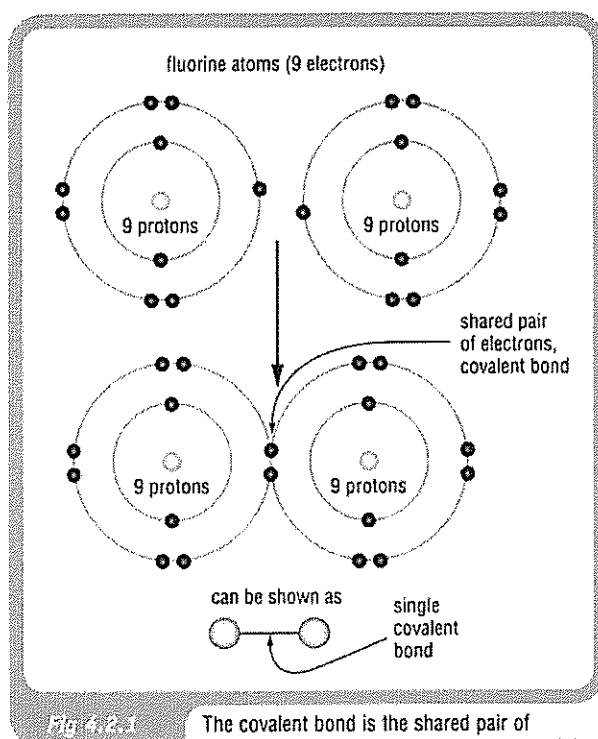
Generally, covalent bonds form between :

- *hydrogen and non-metal elements* on the right hand side of the Periodic Table.
- *non-metal elements* on the right hand side of the Periodic Table.

It is also possible to have *double* or *triple* covalent bonds where two or three pairs of electrons are shared.

e.g. Fluorine molecule  $F_2$ .

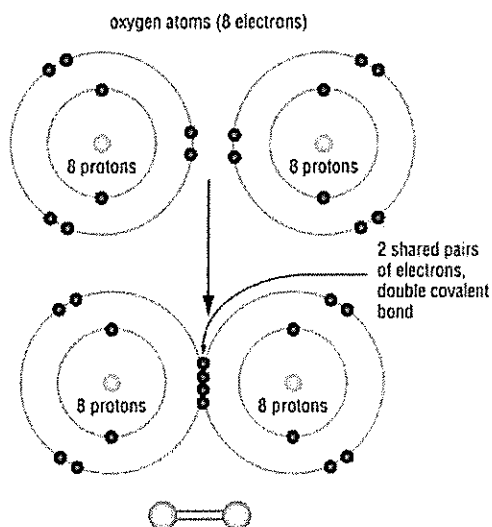
- By sharing one electron from another F atom, both F atoms have the same electron configuration as the nearest inert gas Ne.



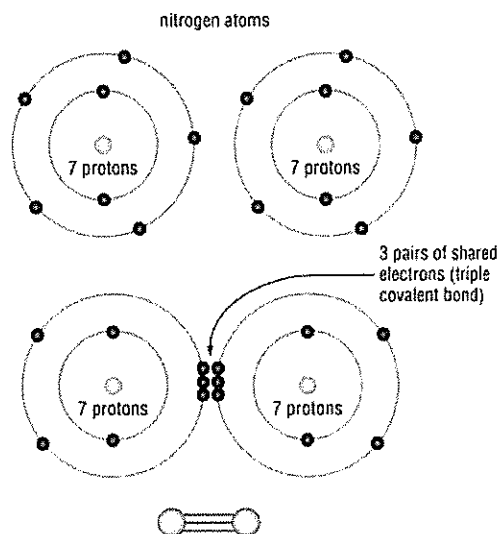
The covalent bond is the shared pair of electrons that occurs between the two nuclei. Note that by sharing a pair of electrons both atoms now have a filled octet of eight electrons in their outer shell—a noble gas configuration.

e.g.

Oxygen gas  $O_2$

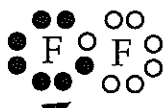


Nitrogen gas  $N_2$



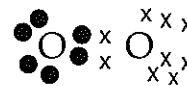
**Note** For covalent substances no square brackets [ ] are used in electron dot diagrams.

e.g. *Single bond*



ELECTRONS ARE  
SHARED, NOT  
DONATED.

*Double bond*

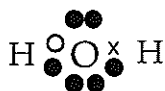


*Triple bond*



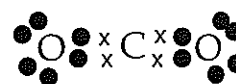
### Other examples

• Water



•

Carbon dioxide



• Draw electron dot diagrams for the following covalent substances.

HCl

$PF_3$

$H_2S$

$CH_3Cl$

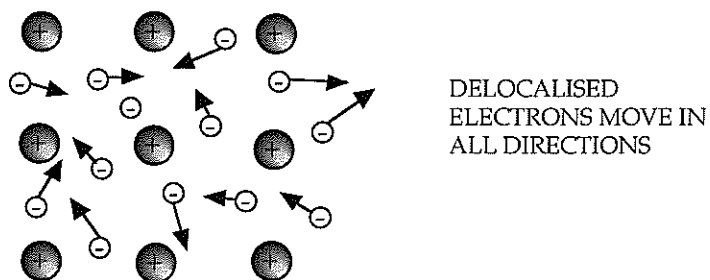


## Metallic Bond

***Metallic Bond:*** due to the electrostatic attraction between delocalised electrons and the positive metallic ions.

In metals, the positive ions are in fixed positions while the delocalised electrons are free to move amongst them.

It is the electrostatic attraction of all the positive ions for the electrons that holds the lattice together.



## WRITING CHEMICAL FORMULAE

Knowledge of the valency of ions allows us to write formulae to represent different chemicals.

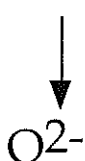
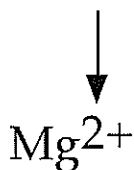
### “RULES”

1. Write down the ions involved and look at the valency of each.
2. If the numbers are the *same*, take one of each ion.
3. If the numbers are *different*, cross the numbers down to the bottom of each ion.

### Examples

magnesium oxide

NUMBERS ARE THE SAME.

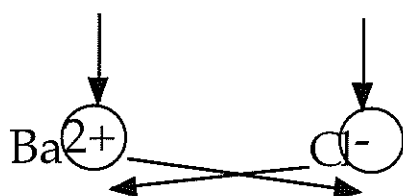


TAKE ONE OF  
EACH ION.



## barium chloride

NUMBERS ARE DIFFERENT.



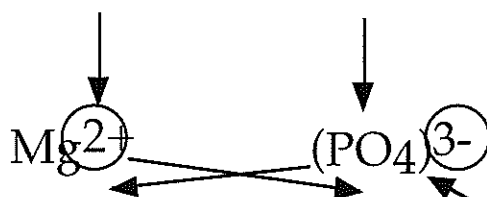
CROSS NUMBERS  
DOWN.



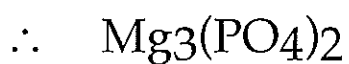
THE "1" DOESN'T HAVE  
TO BE PUT IN.

## magnesium phosphate

NUMBERS ARE DIFFERENT.



CROSS NUMBERS  
DOWN.



USE BRACKETS FOR IONS  
WITH MORE THAN ONE  
ATOM PRESENT.

## Note

The table of ions will be given to you for use in tests and exams - just know how to use them!

# NAMING COMPOUNDS

## Ionic Compounds

Ionic compounds have a *metal element* combined with a *non-metal element* (or elements).

Use the Periodic table to *identify if a metal element is present*. If there is, then the *substance is ionic*.

### Rules for Naming Ionic Compounds

- Name the metal first, then the ion.
- Use the table of common ions (see separate sheet) to help name the compound.

## Examples

| <u>Formula</u>                                                   | <u>Name</u>         | <u>Description</u>   |
|------------------------------------------------------------------|---------------------|----------------------|
| MgCO <sub>3</sub>                                                | magnesium carbonate | 1xMg, 1xC, 3xO atoms |
| Zn(OH) <sub>2</sub>                                              | zinc hydroxide      | 1xZn, 2xO, 2xH atoms |
| <div><div>This multiplies everything in the bracket.</div></div> |                     |                      |
| Cu <sub>3</sub> (PO <sub>4</sub> ) <sub>2</sub>                  | copper phosphate    | 3xCu, 2xP, 8xO atoms |

## Covalent Compounds

Covalent compounds have *all non-metal atoms*.

Use the Periodic Table to *identify if a metal is present*. If *not*, then it is *most likely covalent*.

**Exception** - any substance with the ammonium ion ( $\text{NH}_4^{1+}$ ) in it is ionic.

e.g.     ammonium chloride      $\text{NH}_4\text{Cl}$

### Rules For Naming Covalent Compounds

- Use prefixes such as mono-, di-, tri-, etc.

|      |         |       |         |       |
|------|---------|-------|---------|-------|
| i.e. | mono -  | one   | penta - | five  |
|      | di -    | two   | hexa -  | six   |
|      | tri -   | three | hepta - | seven |
|      | tetra - | four  | octa -  | eight |

## Examples

| <u>Formula</u>                | <u>Name</u>          | <u>Description</u> |
|-------------------------------|----------------------|--------------------|
| SO <sub>3</sub>               | sulphur trioxide     | 1xS, 3xO atoms     |
| CO <sub>2</sub>               | carbon dioxide       | 1xC, 2xO atoms     |
| N <sub>2</sub> O <sub>4</sub> | dinitrogen tetroxide | 2xN, 4xO atoms     |
| SF <sub>6</sub>               | sulphur hexafluoride | 1XS, 6xF atoms     |



**+** Note: You will need to refer to a copy of the periodic table for this worksheet.

Atoms combine with other atoms in order to have complete outermost electron shells. Combined atoms form molecules.

## Part 1: Metals combine with non-metals

### Transferring electrons

**1** Look at the periodic table and complete the following sentences.

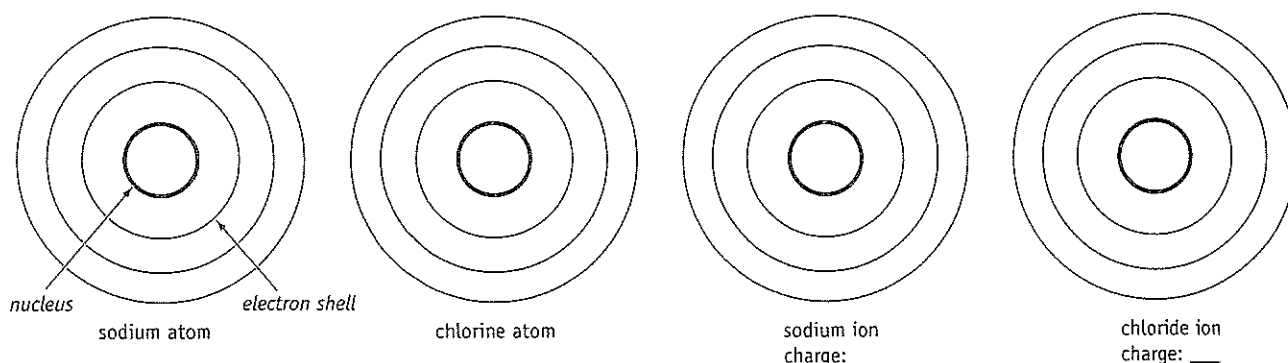
When atoms on the left of the periodic table (ie metals) join with atoms on the right of the periodic table (ie non-metals), electrons are transferred from the metallic atom to the non-metallic atom until the outermost shells are completely full or completely empty.

The metal sodium ( $^{23}_{11}\text{Na}$ ) is in group \_\_\_\_\_ of the periodic table. On the third electron shell sodium has \_\_\_\_\_ electron. If it loses that electron, the new outermost shell is the second shell, which has \_\_\_\_\_ electrons on it, ie it is completely \_\_\_\_\_. In summary, the sodium atom changed into a positive ion when it lost one \_\_\_\_\_.

The non-metal chlorine ( $^{35}_{17}\text{Cl}$ ) is in group \_\_\_\_\_. The outermost electron shell needs to gain \_\_\_\_\_ more electron to make it completely \_\_\_\_\_. In summary, the chlorine atom becomes a negative ion, when it \_\_\_\_\_ an extra electron.

When a sodium atom gives its one lonely electron from the third shell to the chlorine atom, which uses it to fill its outermost shell completely, a molecule has been formed. This molecule, called sodium chloride, consists of two ions: the positive \_\_\_\_\_ ion and the \_\_\_\_\_ chlorine ion. The charges of both ions cancel each other, leaving the \_\_\_\_\_ neutral.

**2** a Complete the diagrams of sodium and chlorine atoms by adding the correct number of protons, neutrons and electrons.



b Indicate in the atoms which electron is transferred during the formation of ions.

c Indicate the charges of the ions by writing 1- and 1+.

d Write the name of the resulting molecule: \_\_\_\_\_

### Ions

When atoms lose or gain electrons, they are no longer neutral and are now called ions. Ions form in order to give atoms completely full outermost electron shells.

A metallic atom from the left side of the periodic table usually loses electrons, thus becoming positively charged. A positive ion has more positive protons than negative electrons. If an atom loses two electrons, the resulting ion has a 2+ charge.

A non-metallic atom from the right side of the periodic table usually gains electrons thus becoming negatively charged. A negative ion has extra electrons compared to its atom. If an atom gains three electrons, its resulting ion has a 3- charge.



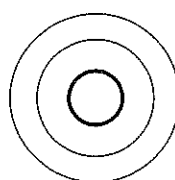
+

- 3** a Name two metals which would form 1+ ions. \_\_\_\_\_  
 b Name two metals which would form 2+ ions. \_\_\_\_\_  
 c Name two non-metals which would form 1- ions. \_\_\_\_\_  
 d Name two non-metals which would form 2- ions. \_\_\_\_\_  
 e Name two non-metals which would form no ions. \_\_\_\_\_  
 f Name two atoms which would form positive or negative ions. \_\_\_\_\_

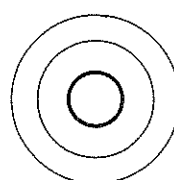
## Ionic molecules

Sometimes the metallic atom wants to lose only one electron in order to make its new outermost shell complete, but the non-metallic atom requires two electrons to fill its outermost shell completely. In this case, two metallic atoms provide the required electrons. The resulting ionic molecule is made up of two positive ions, each with a 1+ charge, and one negative ion with a 2- charge.

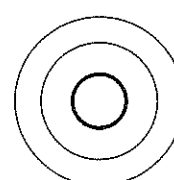
- 4** Complete the atomic diagrams of lithium ( ${}^7_3\text{Li}$ ) and oxygen ( ${}^{16}_8\text{O}$ ) by adding protons, neutrons and electrons.



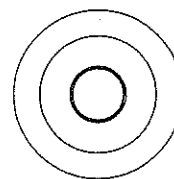
lithium atom



oxygen atom



oxide ion

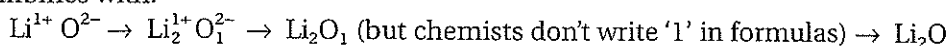


two lithium ions

The molecule formed from the atoms of lithium and oxygen is called lithium oxide and its formula is  $\text{Li}_2\text{O}$ . The metal is named first and remains unchanged. The non-metal comes last and its ending changes to '-ide'.

- 5** a Name the molecules formed from metal magnesium and non-metal fluorine. \_\_\_\_\_  
 b Name the molecules formed from metal aluminium and non-metal sulfur. \_\_\_\_\_

It helps to know the charges of the ions in order to determine the number of atoms in a molecule. The lithium ion has a charge of 1+, while the oxygen ion a charge of 2-. The molecule of lithium oxide is made up of 2 lithium ions and 1 oxygen ion. So the charge of the positive ion tells you how many negative ions it combines with.



- 6** Complete the table about ions and molecules.

| Molecule          | Positive ion | Negative ion | Molecular formula |
|-------------------|--------------|--------------|-------------------|
| calcium bromide   |              |              |                   |
| potassium sulfide |              |              |                   |
| aluminium iodide  |              |              |                   |
| hydrogen oxide    |              |              |                   |

\*Note that hydrogen behaves like a metal here and forms positive ions.

- 7** a State the charge of the complex nitrate ion ( $\text{NO}_3$ ) in the molecule hydrogen nitrate or nitric acid ( $\text{HNO}_3$ ): \_\_\_\_\_  
 b State the charge of the sulfate ion ( $\text{SO}_4$ ) in the molecule hydrogen sulfate or sulfuric acid ( $\text{H}_2\text{SO}_4$ ): \_\_\_\_\_  
 c State the charge of the phosphate ion ( $\text{PO}_4$ ) in hydrogen phosphate or phosphoric acid ( $\text{H}_3\text{PO}_4$ ): \_\_\_\_\_



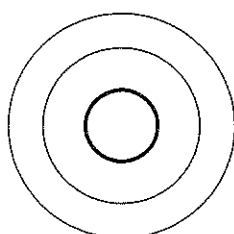


## Part 2: Non-metals combine with non-metals

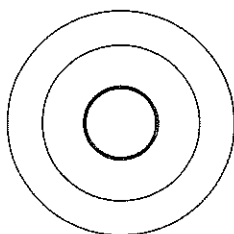
Atoms on the right side of the periodic table (ie non-metals) need only a few electrons to fill their outermost shell completely. When two such atoms want to join to form a molecule, neither one wants to lose electrons and give them to the other. Therefore, ions will not form. When non-metals combine, they share one or two electrons thus ensuring that the outermost shell is completely full.

Fluorine requires one more electron to fill its outermost shell. If it shares one electron with another fluorine atom, its shell is complete. Therefore, two fluorine atoms join to form a fluorine molecule by sharing two electrons, one from each atom. These two electrons make up a single bond. The fluorine molecule can be shown as F-F.

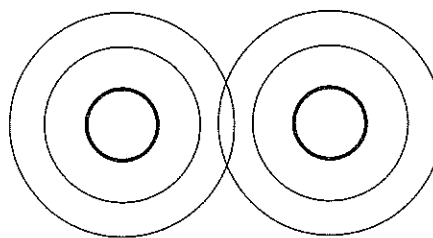
**8** Complete the drawings of the fluorine atoms ( $F_9^{19}$ ) and the fluorine molecule by adding electrons.



fluorine atom



fluorine atom



fluorine molecule  
(showing a single bond)

When two oxygen atoms join to make an oxygen molecule, they form a double bond. Each atom requires two electrons to make its outermost shell complete. Therefore two single bonds (each made up of two electrons) are required. Two single bonds form one double bond. The oxygen molecule is written in its simplified form as  $O=O$ .

**9** Show the bonds for the molecules of the following non-metals:

- chlorine (Cl) \_\_\_\_\_
- sulfur (S) \_\_\_\_\_
- bromine (Br) \_\_\_\_\_
- nitrogen (N) \_\_\_\_\_

When oxygen (requiring two electrons) and fluorine (requiring one electron) join to form fluorine oxide, two fluorine atoms are needed for each oxygen atom. There is one single bond between the oxygen atom and each fluorine atom resulting in the formula  $F-O-F$  or  $OF_2$ .

**10** Show the bonds for the following non-metallic molecules:

- sulfur chloride \_\_\_\_\_
- chlorine fluoride \_\_\_\_\_
- phosphorus hydride \_\_\_\_\_
- nitrogen hydride \_\_\_\_\_

**11** Suggest a reason why helium, neon and argon do not form molecules.

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## TABLE OF COMMON IONS (A VALENCE TABLE)

Below is listed a table of common ions which MUST BE LEARNED. It is impossible to succeed at chemistry unless these symbols are committed to memory and can be recalled for instant use.

| POSITIVE IONS |           |           | NEGATIVE IONS |                    |             |
|---------------|-----------|-----------|---------------|--------------------|-------------|
| +1            | Hydrogen  | $H^+$     | -1            | Chloride           | $Cl^-$      |
|               | Potassium | $K^+$     |               | Iodide             | $I^-$       |
|               | Silver    | $Ag^+$    |               | Fluoride           | $F^-$       |
|               | Sodium    | $Na^+$    |               | Acetate            | $CH_3COO^-$ |
|               | Copper I  | $Cu^+$    |               | Hydrogen carbonate | $HCO_3^-$   |
|               | Ammonium  | $NH_4^+$  |               | Hydroxide          | $OH^-$      |
| +2            | Barium    | $Ba^{2+}$ |               | Nitrate            | $NO_3^-$    |
|               | Calcium   | $Ca^{2+}$ |               | Permanganate       | $MnO_4^-$   |
|               | Copper II | $Cu^{2+}$ | -2            | Oxide              | $O^{2-}$    |
|               | Iron II   | $Fe^{2+}$ |               | Sulfide            | $S^{2-}$    |
|               | Magnesium | $Mg^{2+}$ |               | Carbonate          | $CO_3^{2-}$ |
|               | Lead      | $Pb^{2+}$ |               | Sulfate            | $SO_4^{2-}$ |
|               | Zinc      | $Zn^{2+}$ |               |                    |             |
| +3            | Aluminium | $Al^{3+}$ | -3            | Phosphate          | $PO_4^{3-}$ |
|               | Iron III  | $Fe^{3+}$ |               |                    |             |

## WHAT HAVE YOU LEARNED?

Refer to your TABLE OF COMMON IONS and try the following

- How many electrons does a hydrogen atom lose to become an ion? \_\_\_\_\_
- How many electrons does an oxygen atom gain to become ionized? \_\_\_\_\_
- Does a zinc atom gain or lose electrons to become ionized? \_\_\_\_\_
- If an atom commonly loses three electrons to become ionized. Which element could it belong to? \_\_\_\_\_
- The formula of copper II oxide is  $\text{CuO}$ . How many electrons are transferred from the copper to the oxygen atoms? \_\_\_\_\_
- Explain why *two* sodium ions are needed to combine with *one* sulfate ion to form the ionic solid sodium sulfate,  $\text{Na}_2\text{SO}_4$ .  
\_\_\_\_\_  
\_\_\_\_\_

- Explain why *two* nitrate ions are needed to combine with *one* lead ion to form the ionic solid lead nitrate,  $\text{Pb}(\text{NO}_3)_2$   
\_\_\_\_\_  
\_\_\_\_\_

- Circle the polyatomic ions  $\text{Ba}^{2+}$ ,  $\text{OH}^-$ ,  $\text{CO}_3^{2-}$ ,  $\text{Cu}^+$ ,  $\text{Fe}^{3+}$ ,  $\text{SO}_4^{2-}$ ,  $\text{F}^-$ ,  $\text{CH}_3\text{COO}^-$ ,  $\text{Ag}^+$ ,  $\text{PO}_4^{3-}$ .

- *Without Looking* at your TABLE of COMMON IONS, write in the formula of each of the following ions copper I \_\_\_\_\_, hydroxide \_\_\_\_\_, lead \_\_\_\_\_, calcium \_\_\_\_\_, iodine \_\_\_\_\_, barium \_\_\_\_\_, oxide \_\_\_\_\_, carbonate \_\_\_\_\_.

- Name these compounds

$\text{CuSO}_4$  \_\_\_\_\_

$\text{Al}_2\text{O}_3$  \_\_\_\_\_

$\text{AgCl}$  \_\_\_\_\_

$\text{Na}_2\text{SO}_4$  \_\_\_\_\_

$\text{Ba}(\text{OH})_2$  \_\_\_\_\_

$\text{PbI}_2$  \_\_\_\_\_

$\text{NH}_4\text{NO}_3$  \_\_\_\_\_

$\text{KCl}$  \_\_\_\_\_

$\text{CaCO}_3$  \_\_\_\_\_

$\text{NaCl}$  \_\_\_\_\_

Now try writing formula (remember to balance the ionic charges)

potassium iodide \_\_\_\_\_

potassium permanganate \_\_\_\_\_

copper II sulfate \_\_\_\_\_

lead nitrate \_\_\_\_\_

sodium carbonate \_\_\_\_\_

# WRITE THE CHEMICAL FORMULAE

COMPLETE THE TABLES BELOW BY WRITING THE CORRECT CHEMICAL FORMULA IN EACH SPACE

| 1                | NEGATIVE IONS |                 |               |                 |                    |                    |
|------------------|---------------|-----------------|---------------|-----------------|--------------------|--------------------|
| POSITIVE IONS    | $\text{Cl}^-$ | $\text{O}^{2-}$ | $\text{OH}^-$ | $\text{NO}_3^-$ | $\text{SO}_4^{2-}$ | $\text{CO}_3^{2-}$ |
| $\text{Na}^+$    |               |                 |               |                 |                    |                    |
| $\text{Mg}^{2+}$ |               |                 |               |                 |                    |                    |
| $\text{Zn}^{2+}$ |               |                 |               |                 |                    |                    |
| $\text{Cu}^{2+}$ |               |                 |               |                 |                    |                    |
| $\text{Al}^{3+}$ |               |                 |               |                 |                    |                    |
| $\text{Fe}^{3+}$ |               |                 |               |                 |                    |                    |

| 2             | NEGATIVE IONS |         |           |         |         |           |
|---------------|---------------|---------|-----------|---------|---------|-----------|
| POSITIVE IONS | iodide        | sulfide | hydroxide | nitrate | sulfate | phosphate |
| ammonium      |               |         |           |         |         |           |
| potassium     |               |         |           |         |         |           |
| calcium       |               |         |           |         |         |           |
| tin (II)      |               |         |           |         |         |           |
| lead          |               |         |           |         |         |           |
| iron (III)    |               |         |           |         |         |           |

# Chemical formulas - what a combo!

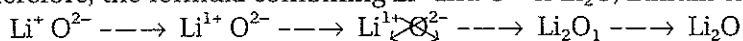
10a



+

**Note:** You will need to refer to a copy of the periodic table for this worksheet.

When you write the formula for a compound, you need to balance its charges by writing numbers as subscripts. For example,  $\text{Li}^+$  has a one plus charge and  $\text{O}^{2-}$  has a two minus charge. When you write a little '2' after the lithium symbol, you are saying that there are two lithiums now. Since each one has a plus one charge, two of them have a two plus charge. And that balances exactly the two minus charge from  $\text{O}^{2-}$ . Therefore, the formula combining  $\text{Li}^+$  and  $\text{O}^{2-}$  is  $\text{Li}_2\text{O}$ , lithium oxide.



**1** Look at the table about chemicals below. Write down each combination in words. Then write down the chemical formula. Some have been done for you already.

|                               | Chloride<br>$\text{Cl}^-$         | Oxide<br>$\text{O}^{2-}$               | Iodide<br>$\text{I}^-$         | Sulfide<br>$\text{S}^{2-}$               | Nitrate<br>$\text{NO}_3^-$         | Sulfate<br>$\text{SO}_4^{2-}$               | Hydroxide<br>$\text{OH}^-$         | Carbonate<br>$\text{CO}_3^{2-}$               |
|-------------------------------|-----------------------------------|----------------------------------------|--------------------------------|------------------------------------------|------------------------------------|---------------------------------------------|------------------------------------|-----------------------------------------------|
| Lithium<br>$\text{Li}^+$      | Lithium chloride<br>$\text{LiCl}$ | Lithium oxide<br>$\text{Li}_2\text{O}$ | Lithium iodide<br>$\text{LiI}$ | Lithium sulfide<br>$\text{Li}_2\text{S}$ | Lithium nitrate<br>$\text{LiNO}_3$ | Lithium sulfate<br>$\text{Li}_2\text{SO}_4$ | Lithium hydroxide<br>$\text{LiOH}$ | Lithium carbonate<br>$\text{Li}_2\text{CO}_3$ |
| Hydrogen<br>$\text{H}^+$      |                                   |                                        | $\text{HI}$                    |                                          |                                    | $\text{H}_2\text{SO}_4$                     |                                    |                                               |
| Magnesium<br>$\text{Mg}^{2+}$ |                                   |                                        | $\text{MgI}_2$                 |                                          | $\text{Mg}(\text{NO}_3)_2$         |                                             |                                    |                                               |
| Sodium<br>$\text{Na}^+$       |                                   | $\text{Na}_2\text{O}$                  |                                |                                          | $\text{NaNO}_3$                    |                                             |                                    |                                               |
| Copper<br>$\text{Cu}^{2+}$    |                                   |                                        |                                | $\text{CuS}$                             |                                    |                                             |                                    | $\text{CuCO}_3$                               |
| Iron<br>$\text{Fe}^{3+}$      | $\text{FeCl}_3$                   | $\text{Fe}_2\text{O}_3$                |                                |                                          |                                    |                                             |                                    |                                               |
| Calcium<br>$\text{Ca}^{2+}$   |                                   |                                        |                                | $\text{CaS}$                             |                                    |                                             | $\text{Ca}(\text{OH})_2$           |                                               |
| Zinc<br>$\text{Zn}^{2+}$      | $\text{ZnCl}_2$                   |                                        |                                |                                          |                                    | $\text{ZnSO}_4$                             |                                    |                                               |
| Aluminium<br>$\text{Al}^{3+}$ |                                   |                                        | $\text{AlI}_3$                 |                                          |                                    |                                             |                                    | $\text{Al}_2(\text{CO}_3)_3$                  |

# Chemical formulas - what a combo! 10b



- 2 Iron compounds can have two different formulas. For example, another formula for iron chloride is  $\text{FeCl}_2$ . This is often called iron II chloride in order to distinguish it from the iron III chloride shown in the table. Write down the formulas of the other iron II compounds from the table.
- 3 Copper compounds can also have two formulas: copper I compounds and copper II compounds.
- State whether copper I or copper II compounds are in the table.
  - Write the formulas for the other copper compounds.
- 4 Use the table above and your periodic table to write the formula for each of these compounds:
- lead nitrate
  - potassium hydroxide
  - silver bromide
  - barium chloride
  - potassium iodide
- 5 Name the following compounds:
- $\text{AgNO}_3$
  - $\text{K}_2\text{S}$
  - $\text{BaS}$
  - $\text{AgI}$
  - $\text{KI}$
- 6 Find the following chemical names in the puzzle below. The 13 remaining letters will spell out an important chemical list.

lithium Li hydrogen magnesium Mg sodium Na copper Cu iron Fe calcium  
zinc Zn aluminium chloride oxide iodide sulfide nitrate sulfate hydroxide  
carbonate lead potassium silver Ag barium bromide

Answer: \_\_\_\_\_

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| C | H | L | O | R | I | D | E | Z | S | O | D | I | U | M |
| A | E | T | A | F | L | U | S | I | M | X | A | R | B | A |
| L | I | T | H | I | U | M | P | N | G | I | E | O | R | G |
| C | O | E | Y | R | L | I | D | C | I | D | L | N | O | N |
| I | D | E | D | I | F | L | U | S | F | E | O | D | M | E |
| U | I | T | R | A | L | U | M | I | N | I | U | M | I | S |
| M | D | C | O | P | P | E | R | E | V | L | I | S | D | I |
| I | E | A | G | M | U | I | S | S | A | T | O | P | E | U |
| C | U | A | E | H | Y | D | R | O | X | I | D | E | C | M |
| L | B | G | N | I | T | R | A | T | E | E | N | A | Z | N |
| B | A | R | I | U | M | E | T | A | N | O | B | R | A | C |

## WORKSHEET: CHEMICAL FORMULAE

1. *Without looking at your notes*, write the symbol and valency for the following ions. The more you can commit to memory, the easier it will be to write formulae correctly.

potassium ion

nitrate ion

magnesium ion

calcium ion

hydroxide ion

sulphate ion

aluminium ion

oxide ion

chloride ion

sodium ion

silver ion

carbonate ion

phosphate ion

sulphide ion

zinc ion

iron (III) ion

bromide ion

copper (II) ion

2. Write the formulae for the following compounds.

sodium chloride

aluminium oxide

calcium chloride

magnesium sulphate

potassium hydroxide

aluminium sulphate

potassium sulphide

copper (I) bromide

magnesium nitrate

calcium phosphate

silver oxide

iron (III) sulphide

aluminium chloride

sodium sulphate

calcium carbonate

zinc nitrate

aluminium sulphide

zinc bromide

sodium carbonate

magnesium phosphate

## WRITING AND BALANCING EQUATIONS

These show the chemical change that occurs in a reaction.

### Rules

1. *Write the equations in words to determine the reactants and products.*

- This step may be omitted.

2. *Write the chemical equation using the correct chemical formulas.*

3. *Subscripts are put after each compound to show the physical state the compound is in.*

(s) = solid      (l) = liquid      (g) = gas      (aq) = in solution

e.g.  $\text{H}_2\text{O}_{(\text{g})}$  = steam     $\text{H}_2\text{O}_{(\text{s})}$  = ice     $\text{NaCl}_{(\text{aq})}$  = salt dissolved in solution.

4. *Count the number and types of atoms on both sides of the equation.*

- Polyatomic ions (like  $\text{SO}_4^{2-}$ ), which do not change from the reactants to the products side, should be considered as a group.

5. *Add numbers in front of the formulas.*

- Make sure that the total number of atoms of reactants is the same as the total number of atoms of products.
- Tick the elements off as you balance them.
- Begin with the element, other than hydrogen or oxygen, that has the greatest number of atoms in any reactant or product.
- Make sure you never change the formula of a chemical substance to balance an equation.

6. *Do a final check.*



## Examples

Balance the following equations.

1. Solid zinc hydroxide + sulphuric acid solution  $\rightarrow$  zinc sulphate solution + water
2. Lead (II) nitrate solution + potassium phosphate solution  $\rightarrow$  solid lead (II) phosphate + potassium nitrate solution
3. heptane (C<sub>7</sub>H<sub>16</sub>) + oxygen gas  $\rightarrow$  carbon dioxide gas + water vapour
4. Copper metal + nitric acid solution  $\rightarrow$  copper (II) nitrate solution + nitrogen dioxide gas + water

## Science understanding

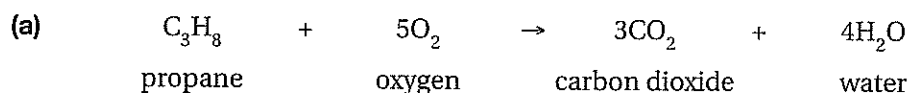
## Logical/Mathematical

Atoms are not created or destroyed in a chemical reaction. Therefore, there must be the same number of atoms in the reactants as in the products of a chemical reaction. This law of science is known as the conservation of mass. It is shown by the use of balanced chemical equations.

**reactant** (n) a substance that reacts with another substance

**product** (n) a substance that is made as a result of a reaction

- 1 **State** the number of each type of atom in the reactants and products of the following equations. **Identify** if the equations are balanced. The first one has been done for you.

**Reactants:**

Carbon (C) = 3

Hydrogen (H) = 8

Oxygen (O) =  $5 \times 2 = 10$

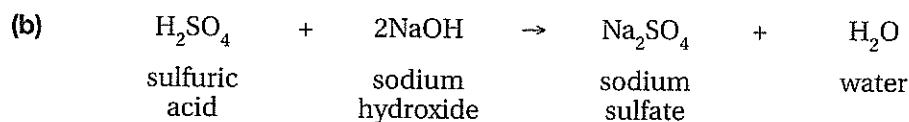
**Products:**

Carbon (C) = 3

Hydrogen (H) =  $4 \times 2 = 8$

Oxygen (O) =  $(3 \times 2) + 4 = 10$

**Balanced/Unbalanced:** balanced

**Reactants:**

Hydrogen (H) = \_\_\_\_\_

Sulfur (S) = \_\_\_\_\_

Oxygen (O) = \_\_\_\_\_

Sodium (Na) = \_\_\_\_\_

**Products:**

Hydrogen (H) = \_\_\_\_\_

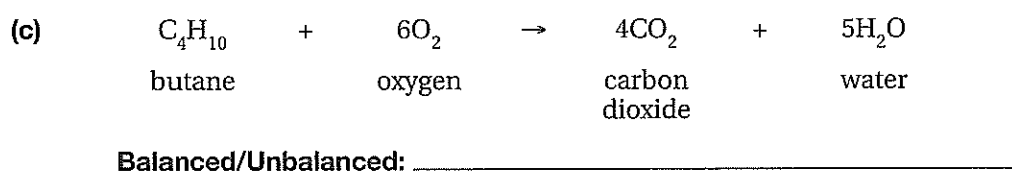
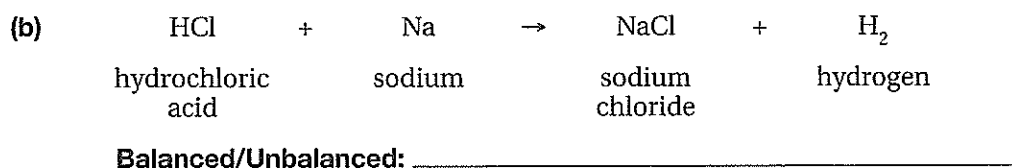
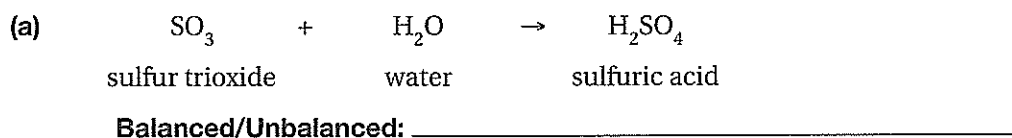
Sulfur (S) = \_\_\_\_\_

Oxygen (O) = \_\_\_\_\_

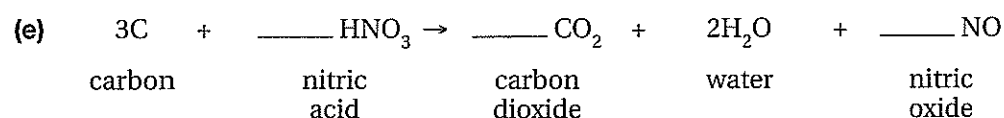
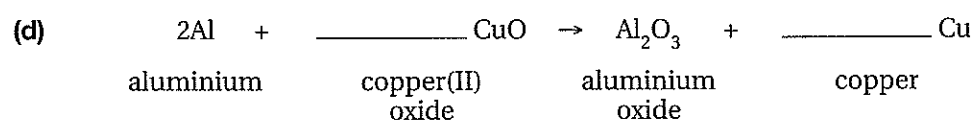
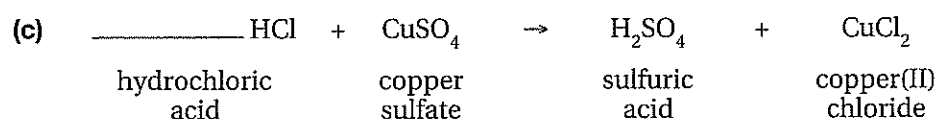
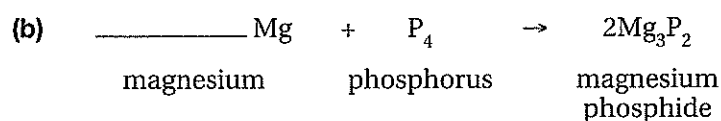
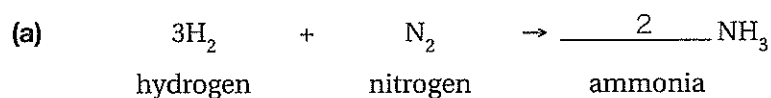
Sodium (Na) = \_\_\_\_\_

**Balanced/Unbalanced:** \_\_\_\_\_

2 State whether the following chemical equations are balanced or unbalanced.



3 Balance the following chemical equations by filling in the blanks. The first one has been done for you.



**EXERCISE 31** Balance the following equations.

1.  $\text{Na} + \text{Cl}_2 \longrightarrow \text{NaCl}$
2.  $\text{CO} + \text{O}_2 \longrightarrow \text{CO}_2$
3.  $\text{Al} + \text{O}_2 \longrightarrow \text{Al}_2\text{O}_3$
4.  $\text{HgO} \longrightarrow \text{Hg} + \text{O}_2$
5.  $\text{Mg} + \text{N}_2 \longrightarrow \text{Mg}_3\text{N}_2$
6.  $\text{P}_4 + \text{O}_2 \longrightarrow \text{P}_4\text{O}_{10}$
7.  $\text{Al} + \text{F}_2 \longrightarrow \text{AlF}_3$
8.  $\text{PbO} + \text{C} \longrightarrow \text{Pb} + \text{CO}_2$
9.  $\text{CH}_4 + \text{O}_2 \longrightarrow \text{CO}_2 + \text{H}_2\text{O}$
10.  $\text{NaHCO}_3 \longrightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2$
11.  $\text{Ca} + \text{H}_2\text{O} \longrightarrow \text{Ca(OH)}_2 + \text{H}_2$
12.  $\text{Fe} + \text{HCl} \longrightarrow \text{FeCl}_2 + \text{H}_2$
13.  $\text{MgCO}_3 + \text{HCl} \longrightarrow \text{MgCl}_2 + \text{H}_2\text{O} + \text{CO}_2$
14.  $\text{NaOH} + \text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
15.  $\text{Mg} + \text{AgNO}_3 \longrightarrow \text{Mg(NO}_3)_2 + \text{Ag}$

**EXERCISE 33** Balance the following equations.

1.  $\text{H}_2\text{S} + \text{O}_2 \longrightarrow \text{H}_2\text{O} + \text{SO}_2$
2.  $\text{Al} + \text{AgNO}_3 \longrightarrow \text{Al(NO}_3)_3 + \text{Ag}$
3.  $\text{NaOH} + \text{H}_3\text{PO}_4 \longrightarrow \text{Na}_3\text{PO}_4 + \text{H}_2\text{O}$
4.  $\text{KClO}_3 \longrightarrow \text{KCl} + \text{O}_2$
5.  $\text{NaOH} + \text{Fe(NO}_3)_2 \longrightarrow \text{Fe(OH)}_2 + \text{NaNO}_3$
6.  $\text{Fe}_2\text{O}_3 + \text{CO} \longrightarrow \text{Fe} + \text{CO}_2$
7.  $\text{Cu} + \text{HNO}_3 \longrightarrow \text{Cu(NO}_3)_2 + \text{H}_2\text{O} + \text{NO}_2$
8.  $\text{NH}_3 + \text{CuO} \longrightarrow \text{Cu} + \text{H}_2\text{O} + \text{N}_2$
9.  $\text{Al}_2(\text{CO}_3)_3 + \text{HCl} \longrightarrow \text{AlCl}_3 + \text{H}_2\text{O} + \text{CO}_2$
10.  $\text{FeS} + \text{O}_2 \longrightarrow \text{Fe}_2\text{O}_3 + \text{SO}_2$

## ACID PROPERTIES

### Acids

- Contain the *element H*.
- Produce  $H^{1+}$  *ions* in solution.
- Have a *sour taste*.
- Turn *blue litmus red*.

e.g.      HCl - hydrochloric acid  
            HNO<sub>3</sub> - nitric acid  
            H<sub>2</sub>SO<sub>4</sub> - sulphuric acid  
            CH<sub>3</sub>COOH - acetic acid

The concentration of acids (or any solution) is given by its *molarity*.

i.e. 2 M H<sub>2</sub>SO<sub>4</sub> is more concentrated than 0.5 M H<sub>2</sub>SO<sub>4</sub>

### Bases

- Any *metal oxide or hydroxide*.
- Produce  $OH^{1-}$  *ions* in solution.
- Have a *soapy feel* and *bitter taste*.
- Turn *red litmus blue*.

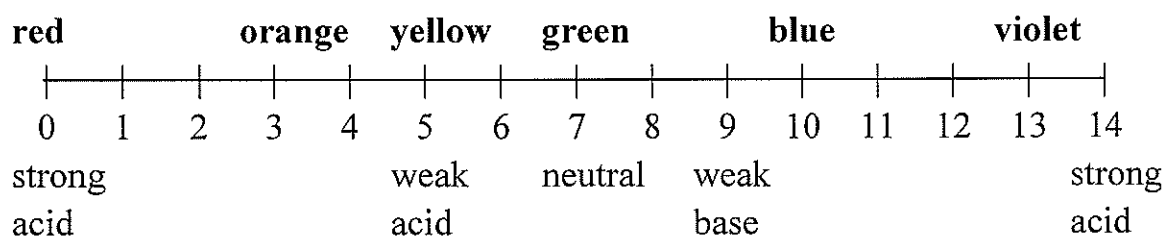
e.g.      NaOH              CaO  
            Mg(OH)<sub>2</sub>        NH<sub>4</sub>OH

### Note

NH<sub>4</sub><sup>1+</sup> behaves as if it is a metal!

## pH SCALE

- Used to determine if a substance is an acid or base.



## REACTIONS

When an acid reacts with another substance, it produces a *salt*.

*Salt*: non-metal part of an acid combined with a metal.

### Examples

HCl - produces *chloride* salts.

e.g. magnesium chloride, sodium chloride, etc.

HNO<sub>3</sub> - produces *nitrate* salts.

e.g. magnesium nitrate, sodium nitrate, etc.

H<sub>2</sub>SO<sub>4</sub> - produces *sulphate* salts.

e.g. magnesium sulphate, sodium sulphate, etc.

CH<sub>3</sub>COOH - produces *acetate* salts.

e.g. magnesium acetate, sodium acetate, etc.

A special reaction of acids is *neutralisation*.

*Neutralisation*: where a base is added to an acid (or vice versa) until the solution contains neither an acid nor a base.

In the case of a large acid spill in the environment, say on a road, emergency service workers will neutralise the acid by adding enough base to neutralise it. Then it can be safely cleaned up.

## ACID REACTIONS

- Only word equations are required for Year 9.

### Reactions with Metals

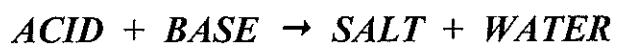


e.g. magnesium + hydrochloric acid → magnesium chloride + hydrogen

zinc + sulphuric acid → zinc sulphate + hydrogen

NOTE - This does not apply to nitric acid. It produces another gas (NO or NO<sub>2</sub>).

### Reactions With Bases



e.g. sodium hydroxide + acetic acid → sodium acetate + water

copper oxide + hydrochloric acid → copper chloride + water

aluminium hydroxide + sulphuric acid → aluminium sulphate + water

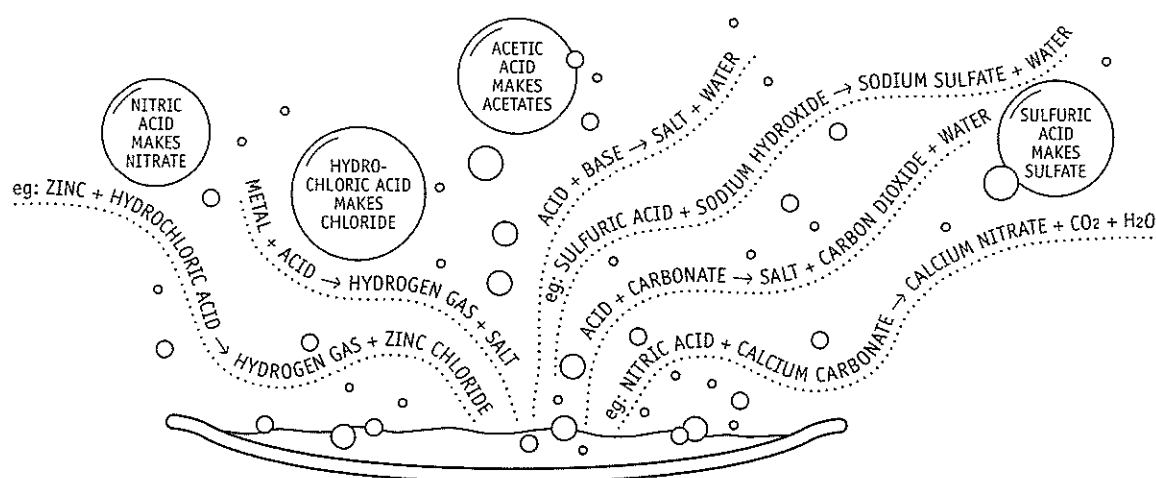
### Reaction With Carbonates / Hydrogencarbonates



e.g. ammonium carbonate + nitric acid → ammonium nitrate + water +  
carbon dioxide

silver hydrogencarbonate + acetic acid → silver acetate + water +  
carbon dioxide

barium carbonate + hydrochloric acid → barium chloride + water +  
carbon dioxide



**1** Complete the following word equations using the rules above to help you. Then draw a circle around all the acids, box the bases, underline the carbonates and highlight all the metals.

- iron + hydrochloric acid  $\rightarrow$  iron chloride + \_\_\_\_\_
- magnesium + hydrochloric acid  $\rightarrow$  \_\_\_\_\_ + hydrogen gas
- zinc + hydrogen sulfate  $\rightarrow$  zinc sulfate + \_\_\_\_\_
- magnesium + sulfuric acid  $\rightarrow$  \_\_\_\_\_ + \_\_\_\_\_
- lead + hydrogen nitrate  $\rightarrow$  \_\_\_\_\_ + \_\_\_\_\_
- zinc + acetic acid  $\rightarrow$  zinc acetate + \_\_\_\_\_
- potassium + acetic acid  $\rightarrow$  \_\_\_\_\_ + \_\_\_\_\_
- sodium hydroxide + nitric acid  $\rightarrow$  \_\_\_\_\_ + \_\_\_\_\_
- magnesium carbonate + sulfuric acid  $\rightarrow$  \_\_\_\_\_ + \_\_\_\_\_ + \_\_\_\_\_
- calcium carbonate + hydrochloric acid  $\rightarrow$  \_\_\_\_\_ + \_\_\_\_\_ + \_\_\_\_\_
- iron carbonate + acetic acid  $\rightarrow$  \_\_\_\_\_ + \_\_\_\_\_ + \_\_\_\_\_
- \_\_\_\_\_ + \_\_\_\_\_  $\rightarrow$  hydrogen gas + sodium chloride
- \_\_\_\_\_ + \_\_\_\_\_  $\rightarrow$  sodium sulfate + water

**2** Write a word equation for the following reactions, then box the reactants and circle the products for each reaction.

- Hydrochloric acid was added to calcium carbonate powder. Bubbles of carbon dioxide formed together with water. The water was evaporated leaving crystals of calcium chloride.  
\_\_\_\_\_
- Sulfuric acid was added to a solution of potassium hydroxide. After the reaction the water that formed was evaporated, leaving behind potassium sulfate.  
\_\_\_\_\_
- A piece of iron was dropped into some nitric acid. Hydrogen gas was produced leaving behind iron nitrate.  
\_\_\_\_\_
- When copper carbonate was added to hydrochloric acid, carbon dioxide and water were produced together with \_\_\_\_\_.  
\_\_\_\_\_
- A small piece of calcium was added to acetic acid producing calcium acetate and bubbles of \_\_\_\_\_.  
\_\_\_\_\_



**YEAR 9 SCIENCE**  
**NATURAL AND PROCESSED MATERIALS**  
**WORKSHEET - ACIDS, BASES AND SALTS**

1. Determine whether the following substances are *ACID (A)*, *BASE (B)* or *SALT (S)*. Write the appropriate letter next to the substance.

- |                     |                          |                                                     |                                       |
|---------------------|--------------------------|-----------------------------------------------------|---------------------------------------|
| (a) NaCl            | (b) MgO                  | (c) zinc hydroxide                                  | (d) HNO <sub>3</sub>                  |
| (e) copper sulphate | (f) copper oxide         | (g) Al <sub>2</sub> (SO <sub>4</sub> ) <sub>3</sub> | (h) H <sub>2</sub> SO <sub>4</sub>    |
| (i) NaOH            | (j) HCl                  | (k) FeCl <sub>3</sub>                               | (l) Ca(NO <sub>3</sub> ) <sub>2</sub> |
| (m) sodium oxide    | (n) CH <sub>3</sub> COOH | (o) potassium acetate                               |                                       |

2. Complete the following general word equations for acid reactions.

ACID + METAL → \_\_\_\_\_ + \_\_\_\_\_

ACID + BASE → \_\_\_\_\_ + \_\_\_\_\_

ACID + CARBONATE → \_\_\_\_\_ + \_\_\_\_\_  
+ \_\_\_\_\_

3. For the following acid reactions, *complete the word equations*. Use the general equations in Q2 above to help you.

(a) magnesium oxide + sulphuric acid →

(b) aluminium + hydrochloric acid →

(c) sodium carbonate + sulphuric acid →

(d) potassium hydroxide + nitric acid →

(e) calcium + acetic acid →

(f) copper oxide + acetic acid →

(g) iron carbonate + hydrochloric acid →

(h) zinc + sulphuric acid →

(i) calcium carbonate + nitric acid →

## PRECIPITATION REACTIONS

When two soluble ionic solutions are mixed it is possible that one of the new combinations of ions produces an *insoluble salt*. These types of reactions are known as *precipitation reactions*.

To predict whether a precipitate will form, a table of *solubility rules* is used.

### SOLUBLE

All **Group I** cations (*lithium, sodium, potassium ions*), *ammonium ion*.

All *nitrates* and *acetates*.

All *chlorides, bromides, iodides* except *silver, lead (II) + mercury ions*.

All *sulphates* except *barium, lead (II) + calcium ions*.

### INSOLUBLE

All *oxides* and *hydroxides* except **Group I** (*lithium, sodium, potassium ions*),  
*ammonium + barium ions*.

All *sulphides* except **Group I** (*lithium, sodium, potassium ions*), *ammonium ion*.

All *carbonates, phosphates* and *sulphites* except **Group I** (*lithium, sodium, potassium ions*), *ammonium ion*.

### Using The Solubility Rules

Barium nitrate and sodium sulphate solutions are mixed. Will a precipitate form?  
If yes, write a molecular and net ionic equation to represent the reaction.

1. **Write down the name for each solution being mixed.** They can both be indicated as (aq) because they are soluble:

barium nitrate (aq) + sodium sulphate (aq)

2. **Look at the two possible new combinations of ions by swapping the names of the metal ions. Check out the solubility table to see if an insoluble substance is produced.**

The solubility rules indicate that, of the two possible products *barium sulphate* and *sodium nitrate*, the barium sulphate is insoluble.

3. **Write out an equation that identifies the insoluble substance formed with an (s).** The other possible combination can be represented with an (aq) to indicate that it remains in solution and is soluble.

barium nitrate (aq) + sodium sulphate (aq) → barium sulphate (s) + sodium nitrate (aq)

## Examples

Write word equations for the following mixtures of solutions. If no precipitate forms, write "no reaction".

- (a) sodium phosphate + aluminium nitrate
- (b) copper (II) chloride + silver nitrate
- (c) ammonium sulphide + potassium bromide
- (d) calcium acetate + sodium carbonate
- (e) potassium hydroxide + lead (II) nitrate
- (f) calcium nitrate + sodium sulphate
- (g) sodium sulphite + potassium phosphate

# To precipitate or not to precipitate

15a



+

When salts are added to water, they often dissolve forming a positive and a negative ion. When solutions of two different salts are mixed, it is possible that one of the positive ions will join with one of the negative ions to form a solid, which will not dissolve. This new solid is called a *precipitate*. Precipitates usually make the solution opaque and colourful. Eventually the precipitate settles to the bottom leaving a transparent solution on top.

Below is a table showing whether mixtures of positive ions (all metals here) and negative ions are soluble (✓) in water or are insoluble (x) in water. If they are insoluble, they form a precipitate.

| Positive ion | Negative ion |          |           |         |         |
|--------------|--------------|----------|-----------|---------|---------|
|              | Carbonate    | Chloride | Hydroxide | Nitrate | Sulfate |
| Barium       | x            | ✓        | ✓         | ✓       | x       |
| Calcium      | x            | ✓        | ✓         | ✓       | x       |
| Copper       | x            | ✓        | x         | ✓       | ✓       |
| Lead         | x            | x        | x         | ✓       | x       |
| Magnesium    | x            | ✓        | x         | ✓       | ✓       |
| Potassium    | ✓            | ✓        | ✓         | ✓       | ✓       |
| Silver       | x            | x        | x         | ✓       | x       |
| Sodium       | ✓            | ✓        | ✓         | ✓       | ✓       |

Below is a table showing the colours of many salts.

| Positive ion | Negative ion |          |           |         |         |
|--------------|--------------|----------|-----------|---------|---------|
|              | Carbonate    | Chloride | Hydroxide | Nitrate | Sulfate |
| Barium       | white        | white    | white     | white   | white   |
| Calcium      | white        | white    | white     | white   | white   |
| Copper       | green        | green    | blue      | blue    | blue    |
| Lead         | white        | white    | white     | white   | white   |
| Magnesium    | white        | white    | white     | white   | white   |
| Potassium    | white        | white    | white     | white   | white   |
| Silver       | white        | white    | brown     | white   | white   |
| Sodium       | white        | white    | white     | white   | white   |

- 1** Write word equations for the reactions, when solutions of the following salts are mixed. Underline the name of the precipitate, if one forms. One example has been done for you.

- sodium carbonate + copper chloride → sodium chloride + copper carbonate
- silver nitrate + potassium chloride → \_\_\_\_\_ + \_\_\_\_\_
- copper nitrate + calcium chloride → \_\_\_\_\_ + \_\_\_\_\_
- barium nitrate + potassium hydroxide → \_\_\_\_\_ + \_\_\_\_\_
- lead nitrate + sodium sulfate → \_\_\_\_\_ + \_\_\_\_\_

- 2** Predict the colour of the precipitate in the following reactions. If no precipitate forms, write 'no precipitate'.

- magnesium nitrate + sodium carbonate → \_\_\_\_\_

# To precipitate or not to precipitate

15b



- + b calcium hydroxide + potassium chloride → \_\_\_\_\_  
 c sodium chloride + silver nitrate → \_\_\_\_\_  
 d copper chloride + potassium carbonate → \_\_\_\_\_  
 e barium nitrate + copper sulfate → \_\_\_\_\_

When formulas are written for a compound which is made up of positive and negative ions, the total charge of the compound has to be zero. The superscript behind the symbol in the table shows you what the charge of the ion is. When you balance the ions in a compound, you write subscripts to indicate how many positive or negative ions there are. Here is an example worked out for you:

$\text{Ba}^{2+}$  and  $\text{Cl}^-$  become  $\text{BaCl}_2$   
 ie the one 2+ ion of barium is balanced by two 1- ions of chloride  
 $\text{Ba}^{2+} \text{Cl}^- \rightarrow \text{Ba}^{2+} \times 2 \text{Cl}^{1-} \rightarrow \text{Ba}_1 \text{Cl}_2 \rightarrow \text{BaCl}_2$

- 3 Write the formulas for the possible compounds in the table below. Some examples have been done for you already.

| Positive ion                  | Negative ion                                 |                           |                                           |                                           |
|-------------------------------|----------------------------------------------|---------------------------|-------------------------------------------|-------------------------------------------|
|                               | Carbonate<br>( $\text{CO}_3$ ) <sup>2-</sup> | Chloride<br>$\text{Cl}^-$ | Hydroxide<br>( $\text{OH}$ ) <sup>-</sup> | Nitrate<br>( $\text{NO}_3$ ) <sup>-</sup> |
| Calcium<br>$\text{Ca}^{2+}$   |                                              | $\text{CaCl}_2$           |                                           |                                           |
| Copper<br>$\text{Cu}^{2+}$    |                                              |                           |                                           | $\text{Cu}(\text{NO}_3)_2$                |
| Magnesium<br>$\text{Mg}^{2+}$ |                                              |                           | $\text{Mg}(\text{OH})_2$                  |                                           |
| Silver<br>$\text{Ag}^+$       | $\text{Ag}_2\text{CO}_3$                     |                           |                                           |                                           |
| Sodium<br>$\text{Na}^+$       |                                              |                           |                                           |                                           |

- 4 Answer the following questions about the molecules in the formula table above. The questions relate to *one* particle of each of the compounds.

- a How many positive ions are there in silver carbonate? \_\_\_\_\_  
 b How many negative ions are there in magnesium hydroxide? \_\_\_\_\_  
 c How many ions are in copper carbonate? \_\_\_\_\_  
 d How many ions are there in sodium chloride? \_\_\_\_\_  
 e How many atoms are in the calcium ion? \_\_\_\_\_  
 f How many atoms are in the carbonate ion? \_\_\_\_\_  
 g How many atoms are there in copper chloride? \_\_\_\_\_  
 h How many atoms are there in silver nitrate? \_\_\_\_\_  
 i Which is a molecule with the greatest number of atoms? \_\_\_\_\_  
 How many atoms are there? \_\_\_\_\_  
 j Which is a molecule with the smallest number of atoms? \_\_\_\_\_  
 How many atoms are there? \_\_\_\_\_

## MOLECULAR AND IONIC EQUATIONS

A chemical equation gives information about the bonds that are broken and formed or, in the case of equations involving ions, which of the ions are involved in the reaction.

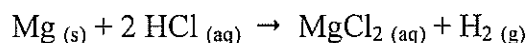
In many equations involving ions, some of the ions are not changed during the chemical reaction. They appear unchanged on both sides of the chemical equation. These ions are called *spectator ions*.

The spectator ions can be left out of the equation as they are not involved in the reaction, leaving what is known as the *net ionic equation*.

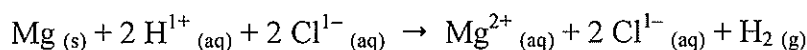
### Examples

What is the net ionic equation for the reaction of hydrochloric acid with magnesium to produce magnesium chloride and hydrogen gas?

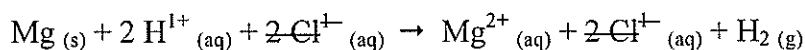
1. *Write a balanced equation in the usual way.*



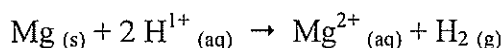
2. *Expand aqueous solutions to clearly identify ions on both sides of the equation.*



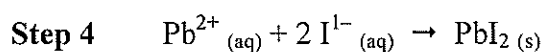
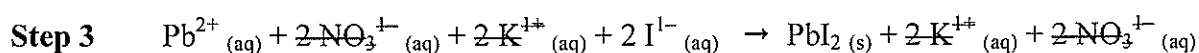
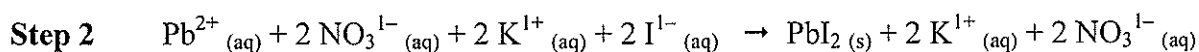
3. *Cross out ions that appear unchanged on both sides of the equation (spectator ions).*



4. *Write out the remainder of the equation - the net ionic equation.*



Lead nitrate solution, when mixed with potassium iodide solution, produces solid lead iodide and a solution of potassium nitrate.



## **SPEEDING UP REACTIONS**

### **State Of Sub-Division (Size)**

- Refers to the size of the pieces of the reactants.
- The smaller the pieces, the faster the rate of reaction.

e.g. Fe in a nail will not burn in a bunsen flame.

Steel wool (which is reasonably open) will burn to some extent in a bunsen flame.

Fe filings are very fine and burn away completely in a bunsen flame.

Smaller pieces increase the effective area available for a reaction to occur.

In the above examples, more  $O_2$  can get to the filings than to the steel wool or nail so they burn very easily.

### **Concentration Of Reactants**

- The greater the concentrations of the reactants, the faster the rate of reaction.

Greater concentrations means that more reactant particles are present.

They can collide more frequently with each other and react to produce product particles.

This is the basis of the Collision Theory.

**Collision Theory:** If particles collide with sufficient energy to break chemical bonds and form new ones, then a chemical reaction will occur.

e.g. More  $\text{H}_2$  gas is produced by adding Zn to 2.0 M HCl (high concentration) rather than 0.5 M HCl (low concentration).

## Temperature

- The higher the temperature, the faster the rate of reaction.

At higher temperatures, the reactant particles are moving faster (i.e. have greater kinetic energy) and collide more often.

Hence bonds are broken more easily and the rate increases.

e.g. A Zn/HCl combination will produce  $\text{H}_2$  at a faster rate if the beaker is heated gently over a bunsen.

## Catalysts

- A catalyst is a substance (or mixture of substances) that increases the rate of a reaction without being used up in the reaction.

e.g.  $\text{O}_2$  is produced very quickly if manganese dioxide ( $\text{MnO}_2$ ) is added as a catalyst to hydrogen peroxide ( $\text{H}_2\text{O}_2$ ).

After the reaction, the  $\text{MnO}_2$  can be seen at the bottom of the reaction vessel - it is unused.



## Science inquiry

Verbal/Linguistic Visual/Spatial Logical/Mathematical

Katrina sets up an experiment to compare how crushing a soluble aspirin tablet affects the rate at which it dissolves. She places a whole aspirin tablet and a crushed aspirin tablet in separate bottles of water and measures the volume of carbon dioxide that is produced over 3 minutes. Her results are summarised in the tables below.

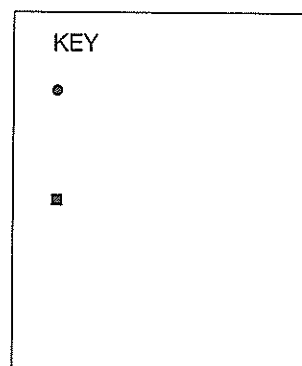
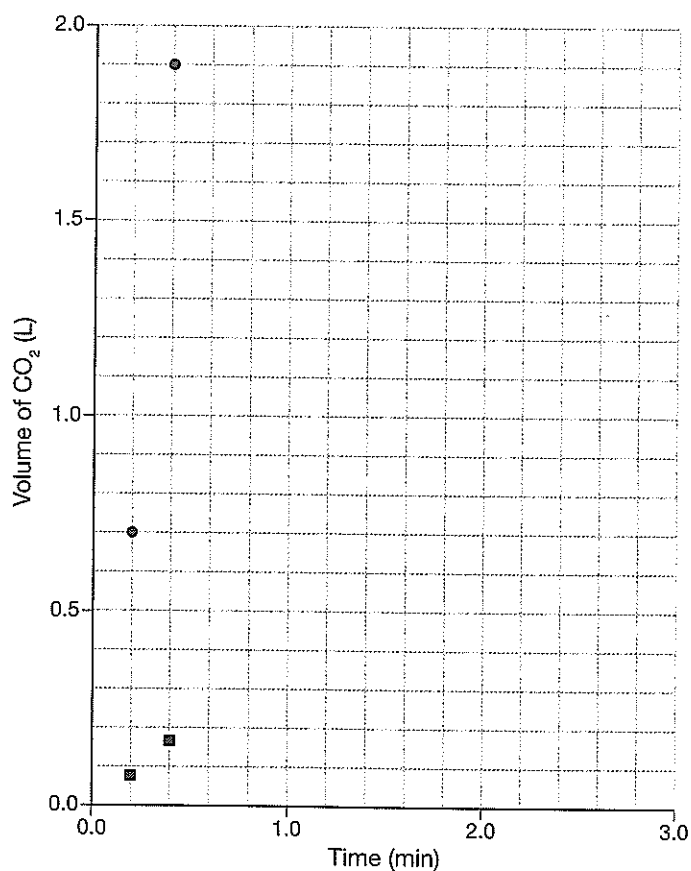
soluble (adj) dissolve in water

|                                       |      |      |      |      |      |      |      |      |
|---------------------------------------|------|------|------|------|------|------|------|------|
| Time (min)                            | 0.0  | 0.2  | 0.4  | 0.6  | 0.8  | 1.0  | 1.2  | 1.4  |
| Volume of CO <sub>2</sub> Control (L) | 0.00 | 0.08 | 0.16 | 0.26 | 0.36 | 0.46 | 0.58 | 0.70 |
| Volume of CO <sub>2</sub> Crushed (L) | 0.00 | 0.70 | 1.90 | 1.90 | 1.90 | 1.90 | 1.90 | 1.90 |

|                                       |      |      |      |      |      |      |      |      |
|---------------------------------------|------|------|------|------|------|------|------|------|
| Time (min)                            | 1.6  | 1.8  | 2.0  | 2.2  | 2.4  | 2.6  | 2.8  | 3.0  |
| Volume of CO <sub>2</sub> Control (L) | 0.84 | 0.98 | 1.14 | 1.31 | 1.49 | 1.69 | 1.90 | 1.90 |
| Volume of CO <sub>2</sub> Crushed (L) | 1.90 | 1.90 | 1.90 | 1.90 | 1.90 | 1.90 | 1.90 | 1.90 |

- 1 Use different symbols to **construct** two graphs from the data in Katrina's experiment on the axes below. Complete the key to show which symbol matches which data. The first two stages have been done for you.





- 2 **Compare** how the whole aspirin tablet and the crushed aspirin tablet reacted when placed in water.

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- 3 **Identify** whether the crushed tablet produced more, less or the same amount of carbon dioxide as the whole tablet. \_\_\_\_\_

- 4 **State** how much carbon dioxide gas was produced by each tablet.

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- 5 From the graph, **calculate** the time it took:

(a) for the whole tablet to react completely \_\_\_\_\_

(b) for the crushed tablet to react completely \_\_\_\_\_

- 6 How much faster did the crushed tablet react compared to the whole tablet? **Calculate** the answer below.

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- 7 A crushed tablet would help to get the aspirin into the bloodstream quickly and relieve pain faster. **Propose** a reason why this soluble aspirin tablet is not taken as a powder or in capsules.

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### HINT

Powders and capsules dissolve very quickly in the stomach. What would happen in your stomach if the aspirin dissolved very quickly?

- 8 Katrina learns that carbon dioxide is produced when sodium bicarbonate ( $\text{NaHCO}_3$ ) and citric acid ( $\text{C}_6\text{H}_8\text{O}_7$ ) in the aspirin tablet react to form sodium citrate ( $\text{NaC}_6\text{H}_7\text{O}_7$ ), carbon dioxide ( $\text{CO}_2$ ) and water ( $\text{H}_2\text{O}$ ). Write the balanced formula equation for this reaction.

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## Science understanding

 Visual/Spatial

Some chemical reactions are very fast while others are very slow. The speed of a chemical reaction is known as the rate of reaction.

**Identify** whether the rate of the chemical reaction is usually increased or decreased in each situation below. **Describe** what is done to control the rate.

|                                                                                                                                                |                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>(a) Baking a cake</p>                                      | <p>(b) Milk going sour</p>                                                      |
| <p>Increased rate <input checked="" type="checkbox"/> Decreased rate <input type="checkbox"/></p> <p>By: <u>increasing the temperature</u></p> | <p>Increased rate <input type="checkbox"/> Decreased rate <input type="checkbox"/></p> <p>By: _____</p>                                                            |
| <p>(c) Food going stale</p>                                  | <p>(d) Capsules to relieve a headache</p>                                      |
| <p>Increased rate <input type="checkbox"/> Decreased rate <input type="checkbox"/></p> <p>By: _____</p>                                        | <p>Increased rate <input type="checkbox"/> Decreased rate <input type="checkbox"/></p> <p>By: _____</p>                                                            |
| <p>(e) Fruit ripening on the way to market</p>              | <p>(f) Respiration during a race</p>                                          |
| <p>Increased rate <input type="checkbox"/> Decreased rate <input type="checkbox"/></p> <p>By: _____</p>                                        | <p>Increased rate <input type="checkbox"/> Decreased rate <input type="checkbox"/></p> <p>By: _____</p>                                                            |
| <p>(g) Starting a camp fire</p>                             | <p>(h) Converting carbon monoxide from a car exhaust into carbon dioxide</p>  |
| <p>Increased rate <input type="checkbox"/> Decreased rate <input type="checkbox"/></p> <p>By: _____</p>                                        | <p>Increased rate <input type="checkbox"/> Decreased rate <input type="checkbox"/></p> <p>By: _____</p>                                                            |

**HINT**

For part (h), refer to page 165 of your student book.

## Science understanding

 Verbal/Linguistic

**Recall** your knowledge of chemical reactions by completing the paragraphs below using words from the box.

|               |               |              |           |                     |               |
|---------------|---------------|--------------|-----------|---------------------|---------------|
| agitation     | area          | balanced     | catalysts | ✓chemical reactions |               |
| combination   | concentration | conservation |           | created             | decomposition |
| destroyed     | electrons     | enzymes      | explosion | insoluble           | oxygen        |
| precipitation | products      | rate         | reactants | rearranged          | redox         |
| rusting       | several       | single       | solutions | temperature         |               |

- (a) Substances can be converted into different substances through chemical reactions. The initial substances are known as the r and the substances that are produced are known as the p. During a chemical reaction, the atoms in the reactants can be r to form the products. However, the atoms can never be c or d. This is known as the law of c of mass. Scientists communicate this by writing b chemical equations.
- (b) Scientists classify chemical reactions into different types. Four types of chemical reactions are d, c, p and r reactions. During a decomposition reaction, a s reactant breaks into several products. During a combination reaction, s reactants combine to form one product. A precipitation reaction occurs when two clear s mix to produce an i solid. In a redox reaction, o atoms or e are transferred from one substance to another.
- (c) The r of a chemical reaction is how fast the reaction takes place. An example of a fast chemical reaction is an e. An example of a slow chemical reaction is r. Nature produces its own chemical helpers, called e, to increase the rate of chemical reactions in your body.
- (d) The rate of chemical reactions can be controlled by:
- controlling the t
  - controlling the c of the reactants
  - controlling the surface a of the reactants
  - stirring or mixing, which is also known as a
  - adding chemical helpers known as c.